Onur Cankur

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Languages: Turkish (native), English (fluent)

Dr. Onur Cankur is a postdoctoral researcher at IT4Innovations, VŠB–Technical University of Ostrava. His research interests span high-performance computing and AI, with a focus on performance analysis, large-scale system telemetry, GPU behaviour, and energy-efficient runtime systems. He received his Ph.D. in Computer Science from the University of Maryland, College Park, where he studied large-scale supercomputer telemetry, run-to-run performance variability, and predictive modelling using GPU, network, and system-level measurements. He has collaborated with researchers at Lawrence Berkeley National Laboratory and Pacific Northwest National Laboratory. He also contributes to open-source software for analysing the performance of parallel applications, including Hatchet and Pipit. Earlier, at Hacettepe University, he worked on GPU-accelerated string-matching algorithms and distributed data storage using blockchain.

EDUCATION

31/08/2020-20/05/2026 **Ph.D., Computer Science,** University of Maryland, College Park, Maryland, USA
- **Dissertation Title:** Telemetry-based Insights and Predictive Analytics for High-performance Computing Systems

03/08/2015-06/07/2020 **B.Sc., Computer Science,** Hacettepe University, Ankara, Turkey

PROFESSIONAL EXPERIENCE

23/07/2026-Present **IT4Innovations National Supercomputing Center,** Postdoctoral Researcher
- Since joining IT4Innovations in July 2026, he has begun extending his expertise towards European HPC systems and runtime energy optimisation.

15/01/2021-20/05/2026 **University of Maryland, College Park,** Graduate Assistant
- Conducted research, supported teaching, and **mentored six undergraduate students** in research projects.

30/05/2023-27/08/2023 **Lawrence Berkeley National Laboratory,** Ph.D. Intern
- Developed models to predict run-to-run performance variability and analysed resource utilisation on the Perlmutter supercomputer at NERSC.

06/06/2022-28/08/2022 **Pacific Northwest National Laboratory,** Ph.D. Intern
- Analysed program memory traces collected with MemGaze.

11/12/2019-01/06/2020 **ASELSAN,** Part-time Software Engineer

17/10/2018-11/12/2019 **Hacettepe University,** Undergraduate Research Assistant
- Parallelized around 20 string-matching algorithms using GPGPU accelerators, and studied big data frameworks and distributed data storage on Blockchain.

22/07/2019-23/08/2019 **ASELSAN,** Summer Intern

18/06/2018-14/08/2018 **Hacettepe University,** Summer Intern

PEER-REVIEWED PUBLICATIONS

Journals

J2. Ozsoy, A., Nazli, M., **Cankur, O.**, & Sahin, C. (2025). CUSMART: effective parallelization of string matching algorithms using GPGPU accelerators. Frontiers of Information Technology & Electronic Engineering, 26(6), 877-895.
Significance: Provides an efficient GPU framework for parallel exact string matching.

J1. Cangir, O. F., **Cankur, O.**, & Ozsoy, A. (2021). A taxonomy for Blockchain based distributed storage technologies. Information processing & management, 58(5), 102627.
Significance: Provides a systematic taxonomy of blockchain-based distributed storage approaches.

Conferences

C3. Cankur, O., Austin, B., Kulkarni, D., & Bhatele, A. (2026, May). Characterizing production gpu workloads using system-wide telemetry data. In 2026 IEEE International Parallel and Distributed Processing Symposium (IPDPS) (pp. 899-912). IEEE.
Significance: Characterises production GPU workloads using large-scale system telemetry from a supercomputer.

C2. Cankur, O., & Bhatele, A. (2022, May). Comparative evaluation of call graph generation by profiling tools. In International Conference on High Performance Computing (pp. 213-232). Cham: Springer International Publishing.
Significance: Evaluates how profiling tools differ in reconstructing parallel-program call graphs.

C1. Nazli, M., **Cankur, O.**, & Ozsoy, A. (2019). A Parallel Comparison of Several String Matching Algorithms Employing Different Strategies. Proceedings Book, 52.
Significance: Compares parallel strategies for accelerating exact string-matching algorithms.

Posters

P1. Cankur, O., Austin, B., & Bhatele, A. (2025, November) Understanding GPU utilization using LDMS data on Perlmutter. In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis, SC '25.

PREPRINTS AND MANUSCRIPTS

1. Cankur, O., Wei, C., Austin, B., Kulkarni, D., & Bhatele, A., Predicting the Performance of Parallel Applications in a Cluster Job Queue. (In preparation)
Significance: Develops predictive methods for estimating the run-to-run performance variability of arbitrary applications running on a production HPC system.

2. Cankur, O., Tomar, A., Nichols, D., Scully-Allison, C., Isaacs, K. E., & Bhatele, A. (2024). Automated Programmatic Performance Analysis of Parallel Programs. arXiv preprint arXiv:2401.13150.
Significance: Enables automated and scalable programmatic analysis of parallel-program profile data.

3. Bhatele, A., Dhakal, R., Movsesyan, A., Ranjan, A., Marry, J., & **Cankur, O.** (2023). Pipit: Enabling programmatic analysis of parallel execution traces. arXiv preprint arXiv:2306.11177.
Significance: Introduces a Python-based tool, Pipit, for programmatic analysis of large parallel execution traces.

CONTRIBUTED SOFTWARE

Hatchet	Enables programmatic analysis of hierarchical performance data.
Pipit	Enables programmatic analysis of large parallel execution traces.
CUSMART	Provides GPU-accelerated implementations of exact string-matching algorithms.

PROFESSIONAL SERVICES

Journal Paper Reviews	TPDS, Frontiers in High Performance Computing
Conference Paper Reviews	IPDPS'27, SCI'27, AgenticAI4HPC'26
External Reviewer	IPDPS'21, IPDPS'22
Student Volunteer	SC'20, SC'21, SC'22, SC'23, SC'25
Organizing Committee	Submissions Co-chair, Charm++ Workshops, 2022

TEACHING EXPERIENCE

UHPC'26 – Introduction to HPC Systems: **19.5 hours of teaching lectures** and **19.5 hours lab sessions**
CMSC131 - Object-Oriented Programming I: Grading and general instructional support
CMSC216 - Introduction to Computer Systems: **28 hours of teaching**; grading; exam proctoring
CMSC416 - Introduction to Parallel Computing: Grading, exam proctoring, review of assessment materials
CMSC616 - Foundations of Parallel Computing: Grading, exam proctoring, review of assessment materials

HONOURS AND AWARDS

Supercomputing Conference Travel Grant	SC'23, SC'25
CS Summer Research Fellowship	University of Maryland, College Park, 2020.
Dean's Fellowship	University of Maryland, College Park, 2020.
Graduated with the fourth-highest GPA	Hacettepe University, 2020.
TÜBİTAK Undergraduate Research Scholarship	Hacettepe University, 2018–2019